Graduate Research Plan Statement
Provably Reliable and Interpretable AI Systems for Therapeutic Enzyme Engineering

My Research Vision

Publicly funded biological data—built on billions in taxpayer investment through NSF, NIH, and other agencies—represents one of science's greatest untapped assets. Yet despite its scale and richness, we lack AI systems capable of **reasoning over this data with rigor, transparency, and reliability**. Today's dominant models are black boxes: they predict, but don't explain; they correlate, but don't guarantee. In therapeutic contexts—where a single error could mean toxic side effects or failed treatments—this opacity is unacceptable. My PhD research will pioneer the first generation of **Verifiable, Interpretable, and Provable AI agents (VIP-AI)** for therapeutic enzyme engineering, closing the critical trust gap between AI and clinical application.

I will integrate cutting-edge neural architectures—protein language models, state-space models (e.g., Mamba), and biophysically constrained attention mechanisms—with formal verification and mathematical priors from structural biology to build models that deliver not just **accuracy**, but **certified safety bounds** and **human-understandable mechanistic explanations**. This work advances core computer science in trustworthy AI while directly enabling safer, more rational design of **enzyme therapeutics—a rapidly growing frontier in precision medicine for cancer, aging, and genetic disease**. By transforming public data into provably reliable knowledge, my research ensures AI earns its place in the lab and the clinic: not as a mysterious oracle, but as a transparent, accountable partner in human health.

Intellectual Merit

Goal 1: Verifiable Neural Architectures for Mutation Effect Prediction

Research Question: How can we design neural architectures with formal verification guarantees that accurately predict mutation effects while respecting fundamental biochemical constraints?

Approach: Current approaches to predicting mutation effects ($\Delta\Delta G$ stability changes, catalytic efficiency) rely on transformers (ESM-2, ProteinMPNN) or physics-based methods (Rosetta, FoldX), but lack formal reliability guarantees and ignore physical constraints. I will develop a novel hybrid architecture combining: (1) *Mamba-based sequence encoders* leveraging linear-time complexity to efficiently process long protein sequences (500+ residues) while maintaining biophysically meaningful hidden states, (2) *constrained attention mechanisms* where QK-V attention weights respect biochemical principles through differentiable projection operators from convex optimization, and (3) *formal verification layers* that provide certified uncertainty bounds using abstract interpretation and SMT solvers, outputting provable guarantees such as "With 95% confidence, the actual $\Delta\Delta G$ lies within [-0.5, 0.3] kcal/mol."

Expected Outcomes: This will produce theoretical guarantees on model reliability quantified through PAC-learning bounds adapted to structured biochemical data, empirically validated architectures on benchmarks (SKEMPI 2.0, Mega-scale), and open-source verification tools for the computational biology community.

Goal 2: Interpretable Reasoning for Enzyme Mechanism Understanding

Research Question: Can we develop AI systems that generate verifiable, mechanistically accurate explanations that structural biologists can validate?

Approach: Understanding *why* mutations improve function is crucial for rational design. I will create neuro-symbolic architectures performing explicit reasoning over enzyme mechanisms through: (1) *structured reasoning with graph neural networks* representing biochemical mechanisms as computational graphs where nodes denote active sites, transition states, and cofactors, with message-passing generating "chains of thought" mirroring expert reasoning, (2) *attention-based mechanism discovery* using mechanistic interpretability to identify which attention heads in protein language models correspond to specific biochemical features (catalytic triads, electrostatic complementarity), and (3) *theorem prover integration* using differentiable theorem provers to verify explanations against biochemical rules and quantum mechanical calculations.

Expected Outcomes: This advances AI interpretability theory while providing practical tools, including formalization of enzyme mechanisms as logical constraints, models generating verifiable explanations with $>80\%$ expert agreement, and case studies designing improved therapeutic enzymes (e.g., L-asparaginase for acute lymphoblastic leukemia).

Goal 3: Certified Safety Guarantees for Therapeutic Enzyme Design

Research Question: How can we provide formal safety certificates for AI-designed therapeutic enzymes, ensuring substrate specificity, stability, and lack of off-target activity?

Approach: Therapeutic enzymes must satisfy stringent safety criteria—enzymes for prodrug activation must be exquisitely substrate-specific, while those degrading toxic metabolites must not cleave essential proteins. Current generative models (ProteinMPNN, ESM-IF) offer no such guarantees. I will develop: (1) *constraint-guided generative models* incorporating hard constraints during sequence generation using projected gradient descent and Lagrangian optimization, provably satisfying constraints like ">100-fold binding selectivity," (2) *uncertainty quantification with certified coverage* through conformal prediction methods providing guaranteed coverage ("With 99% probability, true activity falls within the predicted set"), and (3) *multi-objective optimization with Pareto guarantees* identifying provably Pareto-optimal enzyme variants balancing efficiency, stability, and specificity using Bayesian optimization.

Expected Outcomes: This establishes theoretical foundations for trustworthy AI in protein engineering: formal definitions of safety expressed in temporal logic, algorithms with provable constraint satisfaction guarantees, and experimental validation demonstrating AI-certified enzymes meet predicted specifications.

Broader Impacts

Advancing Human Health and Longevity: This research directly addresses aging and disease challenges. Enzyme replacement therapies for lysosomal storage disorders cost $>\$200{,}000$ per patient annually; improved variants could reduce costs and improve efficacy. My focus on longevity-relevant enzymes (DNA repair glycosylases, NAD+ biosynthesis enzymes, senolytic proteases) accelerates paths from computational prediction to clinical application.

Democratizing Therapeutic Development: Current enzyme engineering requires expensive high-throughput screening accessible only to well-funded institutions. Open-source, reliable AI tools democratize this capability, enabling researchers at resource-limited institutions and developing countries to design therapeutic enzymes computationally. I will release all software with comprehensive documentation and tutorials for structural biologists without extensive ML backgrounds.

Interdisciplinary Education and Mentorship: I am committed to training researchers at the intersection of AI and biology through: (1) workshop series ”Trustworthy AI for Drug Discovery” bringing together ML researchers and biochemists, (2) mentoring undergraduates from underrepresented groups through university STEM diversity initiatives, and (3) curriculum development integrating formal verification into computational biology courses.

Responsible AI Development: By prioritizing interpretability and verification, this work contributes to AI safety and responsible innovation. Methods developed—certified predictions, mechanistic explanations, formal guarantees—transfer to other high-stakes medical AI applications. I will engage with bioethics communities to ensure appropriate safeguards and equitable access.

Conclusion

The convergence of efficient neural architectures, formal verification methods, and computational biology creates an unprecedented opportunity to transform therapeutic enzyme design. My research advances fundamental computer science—developing theory for verifiable learning, interpretable AI, and constrained optimization on structured spaces—while addressing human health challenges. By creating AI systems that are provably reliable and understandable, I aim to establish a new paradigm for trustworthy machine learning in life sciences, embodying NSF’s mission to support research advancing both scientific knowledge and societal well-being.